

1. Create an Incident Response Plan

Rubric Requirements:

- Detect at least 1 security incident
- Containment strategy
- Steps for eradication and recovery
- Identify and explain 1 type of cyber attack

How to demonstrate in your project:

- **Detection:** Use Wazuh alert rules to detect a specific attack, e.g., multiple failed SSH login attempts (brute force attack).
 - **Containment:** Show how alerts could trigger blocking the offending IP or isolating the affected system (you can describe this in the plan; actual network blocking can be simulated).
 - **Eradication & Recovery:** Explain steps like removing malware, resetting passwords, or restoring from backup. You can simulate this by creating a “malicious file” in Parrot and showing its detection and removal.
 - **Attack Type Example:** If using brute force, explain it as a **denial-of-service-style attack on login systems**; if simulating malware, show how Wazuh detects suspicious process executions or file changes.
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2. Develop a Comprehensive Security Policy

Rubric Requirements:

- At least 3 key security rules/guidelines
- Include incident response plan steps
- Explain how policies maintain **CIA Triad** (Confidentiality, Integrity, Availability)

How to demonstrate in your project:

- **Security Rules Example:**
 1. All login attempts must be logged and monitored (Confidentiality + Integrity)
 2. Critical system files must be protected and monitored for changes (Integrity)
 3. All security alerts must be reviewed within 24 hours (Availability + Integrity)
 - **Connection to Wazuh/Parrot:** Show Parrot OS logs collected by Wazuh, custom rules generating alerts, and how analysts would respond according to the policy.
 - **CIA Triad:** Explain how alert monitoring, log protection, and response policies preserve confidentiality, integrity, and availability of systems.
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3. Apply Encryption Techniques

Rubric Requirements:

- Show encrypted text and decrypted plaintext (AES example)
- Show text hashed using MD5 or SHA

How to demonstrate in your project:

- Use Python or terminal commands in Parrot OS:
 - Encrypt a sample text with AES, then decrypt it to show consistency.
 - Hash another sample using MD5 (or SHA1) and show the hash.
 - Document this in a report or a small Python script — even though Wazuh is not handling the encryption, this demonstrates understanding of cryptography as part of cybersecurity basics.
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4. Demonstrate Legal and Ethical Compliance

Rubric Requirements:

- Explain legal/ethical compliance in the incident response plan
- Identify 2 relevant laws/regulations
- Discuss at least 1 ethical consideration
- Explain how your plan upholds these

How to demonstrate in your project:

- **Laws/Regulations Example:** GDPR, HIPAA, or local cybersecurity laws for logging/monitoring
- **Ethical Consideration:** Monitoring user activity without violating privacy; responsible handling of alerts and sensitive logs
- **Connection to Wazuh:** Show that logs are collected securely, alerts are reviewed, and sensitive information is protected according to legal/ethical guidelines.

The screenshot shows a web browser window with the URL `devglan.com/online-tools/aes-encryption-decryption`. The page features a dark-themed header with a navigation bar and a main content area. The main content area is divided into two columns: "AES Encryption" on the left and "AES Decryption" on the right. Both columns have a "Discover Something Every Day" banner at the top. The "AES Encryption" column contains a text input field with the text "My name is Matt and this is my encryption submission.", a dropdown menu for "Select Cipher Mode of Encryption" set to "CBC", a dropdown menu for "Select Padding" set to "PKCS5Padding", a text input field for "Enter IV (Optional)" with the value "Enter initialization vector", a dropdown menu for "Key Size in Bits" set to "128", a text input field for "Enter Secret Key" with the value "aesEncryptionKey", a radio button for "Output Text Format" set to "Base64", and a black "Encrypt" button. Below the button is a text area showing the "AES Encrypted Output" as a long Base64 string. The "AES Decryption" column contains a text input field with the same Base64 string, a dropdown menu for "Select Cipher Mode of Decryption" set to "CBC", a dropdown menu for "Select Padding" set to "PKCS5Padding", a text input field for "Enter IV Used During Encryption(Optional)" with the value "Enter initialization vector", a dropdown menu for "Key Size in Bits" set to "128", a text input field for "Enter Secret Key used for Encryption" with the value "aesEncryptionKey", a radio button for "Output Text Format" set to "Plain-Text", and a black "Decrypt" button. Below the button is a text area showing the "AES Decrypted Output" as the original text "My name is Matt and this is my encryption submission.". On the right side of the page, there is a sidebar with a list of links: "RSA Key Pair Generator", "Online RSA Encrypt Decrypt", "Encrypt Image Online", "Online File Encrypt Decrypt", and "Online Text Encrypt Decrypt". Below the links is a green "Download" button with a download icon and the text "Extension for Mac (Free)". At the bottom of the sidebar, there is a section titled "Only 2 Steps" with a link to "Click 'Download'" and a link to "Add Safe Search".

AES Encryption

Enter Plain Text to Encrypt

My name is Matt and this is my encryption submission.

Select Cipher Mode of Encryption

CBC

Select Padding

PKCS5Padding

Enter IV (Optional)

Enter initialization vector

Key Size in Bits

128

Enter Secret Key

aesEncryptionKey

Output Text Format ☒ Base64 ☐ Hex

Encrypt

AES Encrypted Output

Jk9oRb6Nm7Be0ieEMc+MD9x4qmQc7K9AT8NSq900SZKHwSRAPbFzyVN80qONyh2GXWLdudU91Sk0hrNFH2NuQ==

AES Decryption

AES Encrypted Text

Jk9oRb6Nm7Be0ieEMc+MD9x4qmQc7K9AT8NSq900SZKHwSRAPbFzyVN80qONyh2GXWLdudU91Sk0hrNFH2NuQ==

Select Cipher Mode of Decryption

CBC

Select Padding

PKCS5Padding

Enter IV Used During Encryption(Optional)

Enter initialization vector

Key Size in Bits

128

Enter Secret Key used for Encryption

aesEncryptionKey

Output Text Format ☒ Plain-Text ☐ Base64

Decrypt

AES Decrypted Output

My name is Matt and this is my encryption submission.

Download

Extension for Mac (Free)

Only 2 Steps

Click "Download"

Add Safe Search